Robust Background Feature Extraction Through Homogeneous Region-Based Joint Sparse Representation for Hyperspectral Anomaly Detection

Yixin Yang, Shangzhen Song, Delian Liu, Jianqi Zhang, and Jonathan Cheung-Wai Chan

***Abstract*— Various existing anomaly detection (AD) technologies focus on the background feature extraction and suppression, which serves as a crucial step to extrude anomalies from the hyperspectral imagery (HSI). In this article, motivated by the advantages of the joint sparse representation (JSR) model for adaptive background base selection, a robust background feature extraction method through homogeneous region-based JSR is proposed and used for AD. By segmenting the scene from the spatial domain through an eight-connected region division operation based on the clustering result, a series of nonoverlapping homogeneous regions each sharing a common sparsity pattern are obtained. After discarding small regions, JSR is performed on each region with the dictionary constituted by the overall spectral items in the corresponding cluster. By calculating the usage frequency of dictionary atoms, the most representative background bases describing each background cluster are adaptively selected and then combined into the global background bases. In addition, considering the interference of noise on detection accuracy, an energy deviation-based noise estimation strategy is presented by analyzing the residual obtained from JSR. Finally, the anomaly response of each pixel is measured by comparing its projection energy obtained from the background orthogonal subspace projection with the noise energy in its corresponding region. The proposed method overcomes the shortcomings of traditional neighborhood-based JSR in the common sparsity pattern and anomaly proportion. The spatial characteristics of HSI are fully explored. Furthermore, the interference of noise on detection accuracy is eliminated. Experiments on four HSI data sets demonstrate the superiority of the proposed method.**

***Index Terms*— Anomaly detection (AD), background base selection, hyperspectral imagery (HSI), joint sparse representation (JSR), noise suppression, spectral–spatial characteristics.**

I. Introduction

HYPERSPECTRAL imagery (HSI) captures abundant spectral information with hundreds of contiguous narrow spectral bands, thus providing the possibility to distinguish different ground objects based on their discriminative spectral characteristics [1]. Classification and target detection are the two most typical applications for HSI signal interpretations [2], [3]. Target detection aims to separate specific target pixels from the background and can be regarded as a binary classification problem. However, in real applications, it is sometimes difficult to obtain prior knowledge about the spectral signatures of the targets of interest. Anomaly detection (AD) is an unsupervised target detection problem without any prior spectral knowledge about the targets or background [4], [5]. Since AD does not require spectrum calibration and atmospheric compensation, it has attracted widespread attention in environmental monitoring, mineral exploration, and civil search and rescue [6], [7].

Anomalies in an HSI refer to the unusual observations or objects whose spectra are significantly different from their surrounding background and occupy only a few spatial positions. The common focus of various existing anomaly detectors is the estimation and extraction of background features [8]. Then, anomalies are identified by suppressing the background features in the scene. Integration of spatial information in AD has also become a hot research topic in recent years.

The Reed–Xiaoli (RX) algorithm [9] is the most well-known anomaly detector. It assumes that the background follows a multivariate normal distribution and then detects anomalies by measuring the Mahalanobis distance between each pixel and the background mean. The performance of RX largely depends on an accurate estimation of the background statistics (background mean and covariance matrix) as a form of background representation that is immune to anomaly contamination. To this end, a series of improved versions of RX have been proposed. Local RX (LRX) [10] and cluster-based anomaly detector (CBAD) [11] perform RX in a subset where the pixels are spatially connected or

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spectrally similar, and the background information is, thus, extracted from a relatively homogeneous composition. Kernel RX (KRX) [12] makes use of the nonlinear characteristics between spectral bands by projecting the HSI into a high-dimensional feature space where the Gaussian distribution is approximated. Cluster KRX (CKRX) [13] relieves the tedious calculations in KRX through a clustering-based fast eigendecomposition. However, these anomaly detectors cannot automatically filter anomalies from the background statistics. The low-rank and sparse matrix decomposition (LRaSMD)-based Mahalanobis distance detector (LSMAD) proposed in [14] employs the low-rank component extracted by LRaSMD to estimate the background statistics for RX, but its performance depends largely on the result of LRaSMD and related parameter settings. In addition, all RX-based detectors rely on the assumption of multivariate Gaussian distribution [15], and the subset-based RX faces the challenge of deciding the size of subset because a small size may cause rank deficient when calculating the covariance matrix, while a large size will introduce more anomaly contamination [14].

There are some multivariant outlier detection algorithms dedicated to mitigating the anomaly contamination in background estimation. For instance, the random-selection-based anomaly detector (RSAD) [16] performs a random background selection procedure several times to exclude anomaly pixels based on the estimation of anomaly occurrence probability. The blocked adaptive computationally efficient outlier nominator (BACON) [17] optimizes the background model by using the robust statistics estimated from subsets of the entire HSI scene. However, both the multiple selections in RSAD and the loop iterations in BACON have led to long computing time.

With the development of representation theory in signal and image analysis, representation-based algorithms have received much attention and are applied to AD. They evade the statistical modeling for the complex distribution of background. The sparsity of signals reveals that most of the natural signals can be recovered by only a few base vectors in a low-dimensional subspace, with acceptable performance loss [18]–[20]. Given a suitable background dictionary, the background pixels can be linearly approximated by only a few dictionary atoms, while the anomalies cannot. Thus, the approximation error is used to make a decision on the test pixel being an anomaly or not. Many algorithms have been proposed to construct a background dictionary that contains enough background categories and avoids the anomaly contamination [21], [22].

Joint sparse representation (JSR), as a simultaneous sparse approximation model for several input signals, incorporates the spatial contextual information into the detection procedure of the central test pixel. It is based on the natural redundant prior of the background in HSI [23], [24]. By adaptively filtering out anomalies from the selected background atoms, JSR achieves an automatic background feature extraction. Li *et al.* [23] first introduced JSR into the AD of HSI data by jointly representing all pixels in the local neighborhood captured by sliding hollow dual windows centered on the test pixel through a limited number of common background bases. By assuming that the majority of neighboring pixels belong to the background with similar spectral characteristics, the JSR-based detector considers that these pixels can be simultaneously represented in the same low-dimensional subspace through some common background bases with different compact coefficients [25], [26], that is, they share a common sparsity pattern. The most active atoms in the dictionary are regarded as the most representative background bases for the neighboring pixels. Finally, the central pixel is linearly reconstructed with these extracted local background bases, and the measured error is used for anomaly identification. In this article, we refer to the abovementioned method as the neighborhood-based JSR, which executes JSR in the spatial neighborhood captured by sliding hollow dual windows for each pixel.

However, in real situations, there are two cases that can weaken the assumptions of neighborhood-based JSR and its performance for background feature extraction: 1) the spatial neighbors around the central test pixel do not share a common sparse pattern due to the coverage of multiple material categories and 2) the selected common bases include some anomaly bases because of the occurrence of considerable anomalies in the neighborhood. A few AD algorithms employing JSR have not addressed the abovementioned two shortcomings. For example, in [5], the local JSR is directly used to choose some frequently used atoms as the dictionary atoms for the proposed matrix decomposition-based detector. For the sliding window-based JSR (SWJSR) method in [27], each pixel undergoes a series of different spatial positions with respect to the local neighbors where the JSR is performed. In addition, in [28], the result of spatial JSR performed on the low-dimensional manifold of HSI is combined with the conventional spectral JSR. All of them perform JSR in the local neighborhood, never considering the inherent shortcomings of the neighborhood-based JSR.

To overcome the above two shortcomings, a robust background feature extraction method through homogeneous region-based JSR for AD is proposed in this article. Through performing eight-connected region division [29] on the clustering map, a series of homogeneous regions each with extremely similar spectral characteristics are achieved. Instead of neighborhood-based JSR, in this article, JSR is executed in each homogeneous region after discarding regions with a small number of pixels. With the global background bases selected, the projection responses of pixels to the background orthogonal subspace can be used to yield the decision of whether anomalies exist. Considering the negative interference of noise on detection accuracy, an energy deviation-based noise estimation strategy is applied. The contributions of the proposed method are summarized as follows.

1) Homogeneous region-based JSR effectively overcomes the two essential shortcomings in traditional neighborhood-based JSR through guaranteeing the spectral similarity in each region and discarding small regions.
2) A novel dictionary construction technology for JSR. For one region of interest, the spectral items in regions with the same cluster label as this region are used as its dictionary atoms. Such a dictionary provides a stronger

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spectral approximation to the interested region while also compressing the dictionary subspace.

3) The interference of noise on detection accuracy is fully alleviated by considering the regionwise noise distribution in the scene. The influence of anomalies on noise estimation is also eliminated.

In addition to the traditional neighborhood-based JSR, the AD method by graph pixel selection proposed in [30] also employs the sliding window to divide the scene into a series of overlapping local neighborhoods. Then, vertex- and edge-weighted graphs used for anomaly pixel selection are constructed. Compared with the sliding window strategy, our proposed adaptive region strategy does not need to manually set the window size of the spatial neighborhood. The size and shape of each region are automatically adjusted according to the spectral and spatial characteristics of different materials. Through the eight-connected region division, it provides sufficient consideration on the spatial characteristics of HSIs that have been proved to be beneficial for AD [15], [31]. Moreover, the computational burden is low since the subsequent detection only needs to be executed once for each region. Some existing anomaly detectors are dedicated to suppressing the noise component for undisturbed anomaly recognition. The LRaSMD model [14], [29], [32] is the most typical, which decomposes an HSI into a low-rank matrix, a sparse matrix, and an additive noise matrix. For the hyperspectral AD by fractional Fourier entropy in [33], the fractional Fourier transform (FrFT) is executed to exploit the spectral representation in an intermediate domain between the original reflectance spectrum and its Fourier transform. During FrFT, the noise component is removed. Compared with these detectors estimating the noise component from a global view, our method exploits the regionwise noise energy by measuring the residual of homogeneous region-based JSR, without introducing any other complicated operations or parameters. Since different materials usually exhibit various noise levels, the regionwise noise suppression is more reasonable than globalwise for our method.

The remainder of this article is organized as follows. Section II provides a brief review of the traditional neighborhood-based JSR model for AD and its shortcomings. Section III introduces the proposed method in detail. The related experiments and analysis are presented in Section IV. Finally, Section V concludes this article.

II. Neighborhood-Based JSR for AD

A. Neighborhood-Based JSR Model

Given a background dictionary, the sparse representation (SR)-based anomaly detector considers that a background pixel can be approximated as a linear combination of a few dictionary atoms with the corresponding coefficients where most of the terms are zero, while an anomaly cannot. Therefore, the approximation error can be used to measure the anomaly response. However, such a detection technology only considers the spectral characteristic of the test pixel itself while ignoring its spatial neighboring pixels that usually provide nonignorable spectral correlations.

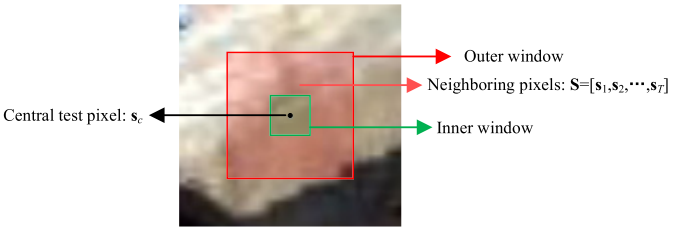


Fig. 1. Neighboring pixels obtained by the sliding hollow dual windows.

JSR is an improved version of SR that incorporates the spatial contextual information when performing detection on the central test pixel. As shown in Fig. 1, local sliding hollow dual windows are employed to traverse each pixel $\mathbf{s}_c \in \mathbb{R}^{B\times 1}$ in an HSI denoted as $\mathbf{X} \in \mathbb{R}^{B\times H}$, where B is the number of bands and H is the number of pixels in $\mathbf{X}$. $\mathbf{s}_c$ is the central test pixel. The outer window marked in red is designed to capture the neighboring pixels (in light pink) represented as $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_T] \in \mathbb{R}^{B\times T}$ around $\mathbf{s}_c$, and the inner window in green is to guard against the interferences from $\mathbf{s}_c$ to $\mathbf{S}$. T is the number of pixels in $\mathbf{S}$.

Based on the highly redundant prior of background and the fact that anomalies occupy only a few positions (low proportion), most of the spectral items in $\mathbf{S}$ usually have similar background spectra. In this case, it is assumed that the background pixels in $\mathbf{S}$ can be simultaneously represented in a low-dimensional subspace spanned by some common background bases, with different compact coefficient vectors. That is, they share a common sparsity pattern. Accordingly, the model of JSR on $\mathbf{S}$ for AD is formulated as

$$\begin{aligned}\mathbf{S} &= [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_T] \\ &= \left[\mathbf{D}\boldsymbol{\alpha}_1 + \boldsymbol{\psi}_1, \mathbf{D}\boldsymbol{\alpha}_2 + \boldsymbol{\psi}_2, \ldots, \mathbf{D}\boldsymbol{\alpha}_T + \boldsymbol{\psi}_T\right] \\ &= \mathbf{D}[\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_T] + \boldsymbol{\Psi} = \mathbf{DA} + \boldsymbol{\Psi} \end{aligned} \tag{1}$$

where $\mathbf{D} \in \mathbb{R}^{B\times H}$ is the dictionary composed by the spectral items of entire HSI, and $\mathbf{A} = [\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_T] \in \mathbb{R}^{H\times T}$ is the set of all coefficient vectors. $\boldsymbol{\Psi} = [\boldsymbol{\psi}_1, \boldsymbol{\psi}_2, \ldots, \boldsymbol{\psi}_T] \in \mathbb{R}^{B\times T}$ is the residual matrix that represents the noise in $\mathbf{S}$. The optimization problem in (1) is converted into

$$\mathbf{A} = \text{argmin}\|\mathbf{DA} - \mathbf{S}\|_F \quad \text{s.t. } \|\mathbf{A}\|_{row,0} \le K_0 \tag{2}$$

where $\|\cdot\|_{row,0}$ is the number of nonzero rows in $\mathbf{A}$, and K_0 is the upper bound of the joint sparsity level. Equation (2) can be solved by a generalized greedy optimization algorithm called simultaneous orthogonal matching pursuit (SOMP), as detailed in [34] and [35].

After solving $\mathbf{A}$, the most active atoms in $\mathbf{D}$ corresponding to the nonzero rows in $\mathbf{A}$ are regarded as the most representative background bases for $\mathbf{S}$. Anomalies in $\mathbf{S}$ are automatically filtered into the residual matrix $\boldsymbol{\Psi} = \mathbf{DA} - \mathbf{S}$. Finally, by linearly reconstructing $\mathbf{s}_c$ with these selected background bases, the error is used to measure the anomaly response of $\mathbf{s}_c$.

Since the abovementioned traditional JSR-based anomaly detector performs JSR in the local neighbors of each pixel in the scene, in this article, it is called the neighborhood-based JSR.

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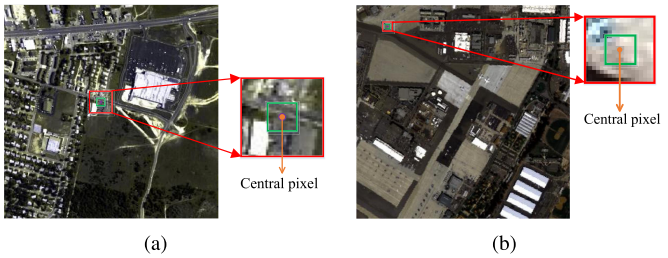


Fig. 2. Some situations that violate the assumptions of neighborhood-based JSR. (a) Complex multiple backgrounds in the neighborhood. (b) Considerable anomaly proportion in the neighborhood.

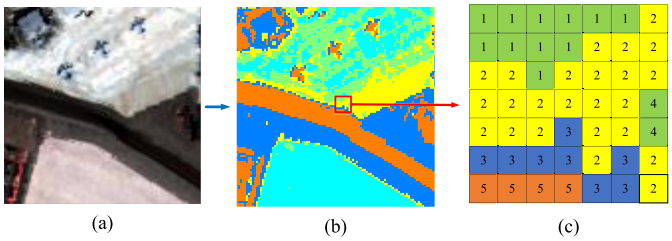


Fig. 3. Schematic illustration of eight-connected region division. (a) False-color image. (b) Clustering map. (c) Eight-connected region map where all pixels with the same label are considered to be in one region.

B. Shortcomings

Neighborhood-based JSR achieves an automatic background feature extraction through the adaptive selection of local background bases. However, its assumptions about the distribution of HSIs are not always applied. In real situations, some local neighborhoods may cover multiple background material categories, especially when the central test pixel is located at the edge of different materials or when the composition of HSI is complex. For example, in the enlarged view (right side) of Fig. 2(a), the neighboring spectra between the red outer window and the green inner window include asphalt roads, buildings, grasslands, and roofs, and they cannot be simply described by some common background bases simultaneously, that is, they do not share a common sparse pattern. As a result, these background bases selected by neighborhood-based JSR cannot characterize sufficient background features and, eventually, lead to false detection when the central test pixel belongs to the background. On the other hand, some anomalies may occupy a large proportion of the neighborhood. An example is given in the enlarged view of Fig. 2(b), where the proportion of the anomalies designated by the light-colored aircraft in the upper left corner is considerable, which goes against the assumption of neighborhood-based JSR on low anomaly proportion. Under such a circumstance, these common bases selected from the dictionary will inevitably involve anomalies, and seriously missed detection will occur as the local background feature extraction has failed.

III. Proposed Method

In this article, in order to effectively eliminate the above-mentioned shortcomings in the traditional neighborhood-based JSR and achieve the accurate detection of anomalies, a robust background feature extraction method through homogeneous region-based JSR is proposed and used for AD. Details are described in the following.

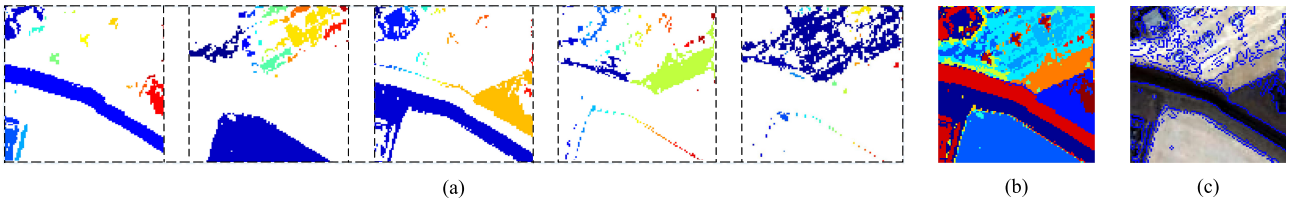
A. Eight-Connected Region Division for Homogeneous Regions

Clustering is one of the most commonly used operations in HSI processing, which aims to assign the same cluster label to pixels with similar spectral characteristics. The K-means++ clustering is first applied on HSI to group similar spectra together. However, in practice, it is generally difficult to achieve a perfect separation between different materials of HSI through a simple clustering operation. Fig. 3(a) shows the false-color image of the San Diego data set (introduced in Section IV-A), and Fig. 3(b) shows the clustering map obtained by K-means++ clustering [36] with cluster number $K = 5$, where each color indicates one cluster label. We can see that, although the three aircraft that are regarded as anomaly objects in this scene are assigned with the same cluster label, the asphalt ground and the roofs in the upper and lower left corners are also wrongly involved. Increasing K cannot solve this dilemma well but makes the same material be further fragmented. Fortunately, we can observe that, although these different materials are grouped into the same cluster, they are separated from each other in the spatial domain. In this case, if the scene is further segmented in the spatial domain based on the clustering map, pixels in each resulted region are largely guaranteed to belong to the same material category. Eight-connected region division [29], [37] is a good choice for spatial image segmentation. Based on the clustering map, it can separate pixels with the same cluster label but spatially disconnected into different regions, according to the spatial aggregation property of homogeneous materials. The details are given as follows.

The eight-connected region of a pixel refers to its eight nearest neighbors in different directions (top, bottom, left, right, top left, bottom right, bottom left, and top right). Eight-connected region division is performed on the clustering map in Fig. 3(b) to further segment the scene according to the distribution characteristic of HSI that the pixels gathered together in the spatial domain are usually belong to the same material. As shown in Fig. 3(c), for an arbitrary pixel $\mathbf{x}$ with cluster label C, all pixels belonging to C in its eight-connected region are marked with the same region label. The same operation is applied to all pixels to extend to the entire spatial scene. In this way, by segmenting the HSI in both of the spectral and spatial domains through the eight-connected region division on the clustering map, we obtain a series of nonoverlapping homogeneous regions where the pixels in any one region have very similar spectral characteristics (belong to the same material). Spectral items in one region are located in a limited subspace spanned by a few corresponding common bases, which is extremely consistent with the assumption of JSR on a common sparsity pattern.

The divided regions for each cluster and the entire scene are shown in Fig. 4(a) and (b), respectively. In any subfigure, each region is painted with a unique color. In addition, the edges of regions are illustrated in Fig. 4(c). Due to display limitations,

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(a) (b) (c)

Fig. 4. Nonoverlapping homogeneous regions. (a) For each cluster. (b) For entire scene. (c) Regional edges.

some regions in Fig. 4(b) look similar in color but actually carry different region labels, such as these three aircraft.

B. Homogeneous Region-Based JSR

For each resulted homogeneous region, JSR is executed with the corresponding dictionary constituted by the overall spectral items in those regions having the same cluster label as this region. Specifically, assume that the set of regions belonging to cluster C is denoted as $\mathbf{R}^C = \{\mathbf{R}_1^C, \mathbf{R}_2^C, \ldots, \mathbf{R}_i^C, \ldots, \mathbf{R}_{N^C}^C\}$, where i the region index and N^C is the number of regions. For region $\mathbf{R}_i^C$ $(i = 1, \ldots, N^C)$, its dictionary for JSR is constructed by $\mathbf{D}^C = \{\mathbf{D}_1^C, \mathbf{D}_2^C, \ldots, \mathbf{D}_i^C, \ldots, \mathbf{D}_{N^C}^C\}$, where $\mathbf{D}_s^C$ $(s = 1, \ldots, N^C, s \neq i)$ collects the spectral items in region $\mathbf{R}_s^C$ and each spectral atom in $\mathbf{D}_i^C$ is a column vector with all zero elements ($\mathbf{0}$-column vector). The formulation of JSR on $\mathbf{R}_i^C$ with $\mathbf{D}^C$ is given as

$$\mathbf{A}_i^C = \text{argmin}\|\mathbf{D}^C\mathbf{A}_i^C - \mathbf{R}_i^C\|_F \quad \text{s.t. } \|\mathbf{A}_i^C\|_{row,0} \leq K_0 \tag{3}$$

where $\mathbf{A}_i^C$ is the set of all coefficient vectors.

It is worth noting that the anomaly regions composed of spatially aggregated anomaly pixels should have been excluded from participating in JSR; otherwise, the anomaly related bases will be involved in the subsequent background base selection and further lead to a contaminated background feature extraction. Unlike background materials that are evenly distributed over a large region, the anomaly targets in HSIs usually occupy a small number of aggregated pixels, which means that the small regions with only a few pixels are very likely to correspond to the anomaly targets in the scene. In this case, directly discarding small regions can effectively avoid the negative interference caused by anomaly pixels. The threshold of pixel number used to define small regions is set as N, which depends on the size of anomaly objects in HSIs and can be effective estimated according to the parameter analysis in Section IV-E.

Through solving the homogeneous region-based JSR formulated as (3), $\mathbf{A}_i^C$ is obtained where each row records the coefficients of the corresponding dictionary atom in $\mathbf{D}^C$ used to represent all spectral items in $\mathbf{R}_i^C$. The positions of nonzero rows in $\mathbf{A}_i^C$ indicate the selected common background bases from $\mathbf{D}^C$. Setting the spectral atoms in $\mathbf{D}_i^C$ as $\mathbf{0}$-column vectors are to avoid the trivial solutions of (3); otherwise, the problem of minimum distance classification phenomenon (the l_0-norm of the coefficient vector is 1) [27] will be introduced. Compared with a dictionary formed by the entire HSI in traditional neighborhood-based JSR, the atoms in our dictionary provide a stronger spectral approximation to the interested region, and the computational complexity of JSR is significantly reduced by compressing the dictionary subspace.

After discarding small regions, although a few scattered anomaly pixels may still exist in some remaining regions, they do not weaken the performance of (3). Such a low proportion of anomalies is automatically filtered into the residual matrix so that the structure of $\mathbf{A}_i^C$ will not be affected.

Then, the homogeneous region-based JSR is performed on all N^C regions in $\mathbf{R}^C$ with respect to the dictionary $\mathbf{D}^C$. N^C coefficient matrices $\Omega = \{\mathbf{A}_1^C, \mathbf{A}_2^C, \ldots, \mathbf{A}_i^C, \ldots, \mathbf{A}_{N^C}^C\}$ are achieved. We focus on Ω, and for each dictionary atom, its usage frequency is measured from Ω to characterize how often it participates in representing regions in $\mathbf{R}^C$, that is, how important it is. Specifically, the usage frequency of a dictionary atom is defined as the number of times it is selected as one of the common background bases to represent the regions in $\mathbf{R}^C$. The usage frequencies of all dictionary atoms in $\mathbf{D}^C$ are measured as

$$F^C = \sum_{i=1}^{N^C} I_S\left(\sum_{p=1}^{M_i^C} |\mathbf{A}_{i,p}^C|\right) \tag{4}$$

where $\mathbf{A}_{i,p}^C$ is the coefficient vector corresponding to the pth column of $\mathbf{A}_i^C$, and M_i^C is the number of pixels in $\mathbf{R}_i^C$. $I_S(\cdot)$ is the indicator function of $S = \{x | x \in \mathbb{R}, x \neq 0\}$, which is used to identify nonzero elements and set them as 1. Items in F^C record the usage frequency of each atom in $\mathbf{D}^C$ and are then arranged in descending order. An intuitive diagram is shown in Fig. 5. The first L most frequently used atoms are treated as the most representative background bases for cluster C. In particular, when there is only one region for the current cluster after discarding small regions, the homogenous region-based JSR formulated in (3) cannot be executed due to invalid dictionary construction. Since all spectral items in this cluster are spatially aggregated and occupy a spatial size that is not small, they are likely to correspond to the background material (anomaly objects in HSIs usually occupy very few pixels and are discretely distributed). In this case, we randomly select some spectral items from this region, which can be regarded as the most representative background bases for the current cluster.

The selected background bases from all clusters are sequentially arranged as $\mathbf{D} = \{\mathbf{B}_1, \mathbf{B}_2, \ldots, \mathbf{B}_K\}$ and constitute the global background bases of the entire HSI scene.

C. AD With Energy Deviation-Based Noise Estimation

With the global background bases $\mathbf{D}$, the anomaly degree of an arbitrary pixel $\mathbf{x}$ in the scene can be judged by calculating

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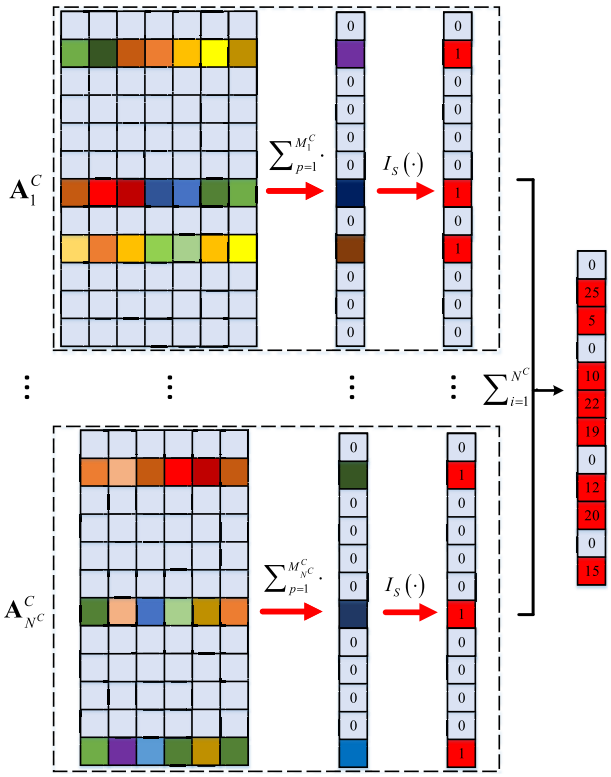


Fig. 5. Calculation of the usage frequency of dictionary atoms.

its projection response into the global background orthogonal subspace [38], [39]. The projection operator is defined as $\mathbf{P}_\mathbf{D}^\perp = \mathbf{I} - \mathbf{D}\mathbf{D}^\#$, where $\mathbf{D}^\# = (\mathbf{D}^T\mathbf{D})^{-1}\mathbf{D}^T$ is the pseudoinverse matrix of $\mathbf{D}$. $\mathbf{x}$ can be denoted as $\mathbf{x} = \mathbf{x}_A + \mathbf{x}_B + \mathbf{x}_N = \mathbf{d}\alpha + \mathbf{D}\boldsymbol{\gamma} + \mathbf{x}_N$, where $\mathbf{x}_A = \mathbf{d}\alpha$, $\mathbf{x}_B = \mathbf{D}\boldsymbol{\gamma}$, and $\mathbf{x}_N$ refer to the background, anomaly, and noise components in $\mathbf{x}$, respectively [32]. $\mathbf{d}$ is the anomaly spectrum, α is the abundance fraction of $\mathbf{d}$ in $\mathbf{x}$, $\mathbf{D}$ is the set of global background bases that we obtained, and $\boldsymbol{\gamma}$ is the abundance vector where each element represents the abundance fraction of the corresponding background base in $\mathbf{D}$. Then, the projection response of $\mathbf{x}$ is given as

$$\begin{aligned}\mathbf{x}_\epsilon &= \mathbf{P}_\mathbf{D}^\perp \cdot \mathbf{x} = \mathbf{P}_\mathbf{D}^\perp \cdot (\mathbf{x}_A + \mathbf{x}_B + \mathbf{x}_N) \\ &= \mathbf{P}_\mathbf{D}^\perp \cdot \mathbf{d}\alpha + \mathbf{P}_\mathbf{D}^\perp \cdot \mathbf{D}\boldsymbol{\gamma} + \mathbf{P}_\mathbf{D}^\perp \cdot \mathbf{x}_N \\ &= \mathbf{P}_\mathbf{D}^\perp \cdot \mathbf{x}_A + \mathbf{P}_\mathbf{D}^\perp \cdot \mathbf{x}_N \end{aligned} \tag{5}$$

where $\mathbf{P}_\mathbf{D}^\perp \cdot \mathbf{D} = \mathbf{0}$ ($\mathbf{0}$ matrix). Through projection, the background components in the scene are fully suppressed, while the anomalies are extruded into $\mathbf{x}_\epsilon$. The anomaly response of $\mathbf{x}$ can be calculated as the l_2-norm of $\mathbf{x}_\epsilon$, and a larger value generally means a higher anomaly probability.

However, as shown in (5), the noise $\mathbf{x}_N$ inevitably causes a negative interference on the detection accuracy of $\mathbf{x}$. A noisy pixel or a background pixel with heavy noise can also generate a large response through (5), which will easily be confused as an anomaly. Accordingly, it is obvious that AD will be benefited by eliminating the noise information from the detection process of pixels [40], [41]. Fortunately, the homogeneous region-based JSR in Section III-B can provide an effective estimation of the noise energy in each region, as follows.

Algorithm 1 HR-JSR for Hyperspectral AD

Input: 1) HSI data $\mathbf{X}$; 2) number of clustering centroids: K; 3) threshold of the pixel number used to define small regions: N; 4) upper bound of the joint sparsity level: K_0; 4) percentage of the selected background bases for each cluster: L;

1. Perform K-means++ clustering and then segment a series of homogeneous eight-connected regions based on the clustering map.
2. Discard small regions and then execute the homogeneous region-based JSR with respect to the dictionary constituted by the overall spectral items in the corresponding cluster.
3. a) Select the most representative background bases for each cluster, and then merge them into the global background bases.

   b) Measure the noise energy of each region with the energy deviation-based noise estimation strategy.
4. Project each pixel into the global background orthogonal subspace and calculate its projection response.
5. Detect anomalies by comparing the projection energy obtained from step 4 and the noise energy in the corresponding region measured in step 3b.

Output: Anomaly detection map of $\mathbf{X}$.

After solving $\mathbf{A}_i^C$ in (3), the residual matrix is obtained as $\boldsymbol{\Psi}_i^C = \mathbf{D}^C\mathbf{A}_i^C - \mathbf{R}_i^C$, which reflects the noise information in $\mathbf{R}_i^C$. In this case, the regionwise noise is measured and used to alleviate the influence of noise in (5). However, a few scattered anomaly pixels may also have been filtered into the residual matrix, which will affect the accurate measurement of the regionwise noise. Considering that the anomalies in the scene usually have significantly higher spectral energy than noises, an energy deviation-based method [42] is presented to eliminate the anomalies in the residual matrix for an accurate noise estimation. The spectral energy of one pixel in HSI refers to the square of the l_2-norm of its spectral vector. Specifically, for region $\mathbf{R}_i^C$, the energy of each pixel in $\boldsymbol{\Psi}_i^C$ is first measured. Then, with the average energy $\boldsymbol{\mu}_i^C$ and the standard energy deviation $\boldsymbol{\sigma}_i^C$, the energy upper limit of $\boldsymbol{\Psi}_i^C$ is defined as

$$E_{\max_i}{}^C = \boldsymbol{\mu}_i^C + 2 \cdot \boldsymbol{\sigma}_i^C. \tag{6}$$

$E_{\max_i}{}^C$ is set as an energy value twice $\boldsymbol{\Psi}_i^C$ from $\boldsymbol{\mu}_i^C$. Pixels with energy values larger than $E_{\max_i}{}^C$ are removed from $\boldsymbol{\Psi}_i^C$, and then, the average energy of remaining pixels in $\boldsymbol{\Psi}_i^C$ is naturally regarded as the noise energy of $\mathbf{R}_i^C$, denoted as $\hat{\boldsymbol{\mu}}_i^C$.

Theoretically, our proposed regionwise noise estimation is more reasonable than the globalwise because different materials in an HSI usually exhibit different noise levels. The final anomaly response of $\mathbf{x}$ is measured by comparing its projection energy into the global background orthogonal subspace with the noise energy in its corresponding region, as follows:

$$w(x) = \frac{\|\mathbf{x}_\epsilon\|_2}{\hat{\boldsymbol{\mu}}_i^C}. \tag{7}$$

The schematic illustration of the proposed method is drawn in Fig. 6, which is called the robust background feature

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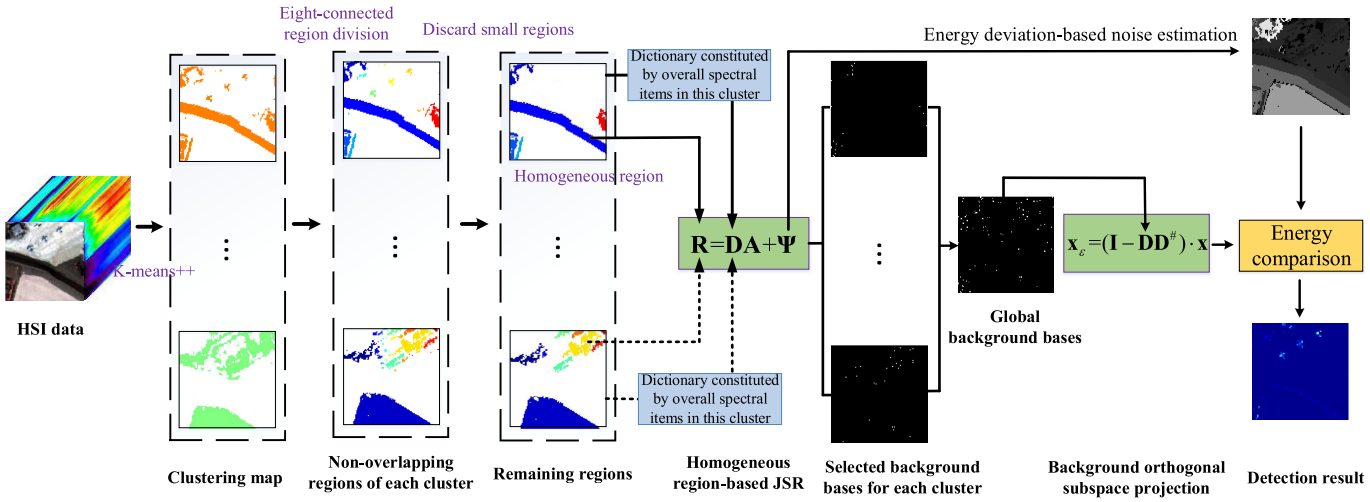


Fig. 6. Schematic illustration of HR-JSR for hyperspectral AD.

extraction through homogeneous region-based JSR for hyperspectral AD (HR-JSR). In addition, the main procedures are summarized in Algorithm 1.

IV. Experiments and Analysis

In this section, the effectiveness and superiority of our proposed HR-JSR are demonstrated with experiments on one synthetic and three real HSI data sets. Seven state-of-the-art anomaly detectors are used as the benchmarks to demonstrate the superiority of HR-JSR for background feature extraction and AD. Color detection map, receiver operating characteristic (ROC) curve [43], [44], area under curve (AUC) [45], and background-anomaly separation map [46] are used as the performance evaluation indicators. ROC plots the curve relationship between the false alarm rate (FAR) and the detection rate (DR) under a series of continuous thresholds. AUC refers to the area under the ROC curve. In the background-anomaly separation map, all detection values are normalized to 0-1, and then, a green box and a red box are assigned for each detector to denote the distribution of background and AD values, respectively. For each box, the central mark indicates the median, and the top and bottom edges represent the lower and upper quartiles, respectively. The whisker is the most extreme value that extends from the end of the box to 1.5 times the interquartile range. Accordingly, a greater distance between two boxes generally means a better performance of the corresponding detector.

In addition, in order to clearly reveal the procedures of HR-JSR in detail, the results of some staged steps for each data set are visualized and analyzed. Finally, the robustness of the performance of HR-JSR to related parameters is evaluated. All experiments are performed with MATLAB 2016a on a personal laptop equipped with an Intel Core i5 2.40-GHz central processing unit, 8-GB memory, and 64-bit Windows 10.

A. Data Set Description

The first data set is a synthetic scene generated from a real HSI data set, which is collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor from the San Diego airport area, San Diego, CA, USA [22]. Its spatial resolution is 3.5 m per pixel, and it has 224 spectral bands spanning the wavelengths from 370 to 2510 nm. After eliminating the bands with low signal-to-noise ratio (SNR) and water vapor absorption, 189 bands are retained. The whole scene contains 400 × 400 pixels, and a subimage with 100 × 100 pixels is intercepted to form the synthetic data set. The anomalies are simulated by the target implantation method [47]. Based on the linear mixing model (LMM), a synthetic subpixel anomaly with spectrum $\mathbf{x}$ and specified abundance fraction f is generated by fractionally implanting the desired anomaly with spectrum $\mathbf{t}$ in a given background pixel with spectrum $\mathbf{b}$ [48], as follows:

$$\mathbf{x} = f \cdot \mathbf{t} + (1 - f) \cdot \mathbf{b} \tag{8}$$

where $\mathbf{t}$ corresponds to the spectrum of an aircraft, which is selected from the middle left of the whole scene; 18 anomaly objects are generated, distributed in three rows and six columns. The size of anomalies from top to bottom in each column is 1 × 1, 2 × 2, and 3 × 3, respectively. In each row, f is 0.1, 0.2, 0.4, 0.6, 0.8, and 1.0 from left to right. The false-color images (generated from bands 52, 31, and 20) of the whole scene and the detection scene are shown in Fig. 7(a). The ground-truth map is illustrated in Fig. 7(b).

The first real-world data set also comes from the San Diego airport image. It is intercepted from the top-left of the whole scene and contains 100 × 100 pixels. Three aircraft occupying 58 pixels are regarded as the anomaly targets in our experiments. The false-color image (bands 52, 31, and 20) and the ground-truth map are exhibited in Fig. 8(a) and (b), respectively.

The second real data set captures a suburban residential area with the Hyperspectral Digital Imagery Collection Experiment (HYDICE) remote sensor. The whole scene has 307 × 307 pixels with a spatial resolution of 10nm and 210 spectral bands ranging from 450 to 2500 nm [15]; 160 bands are remained after excluding the low SNR and

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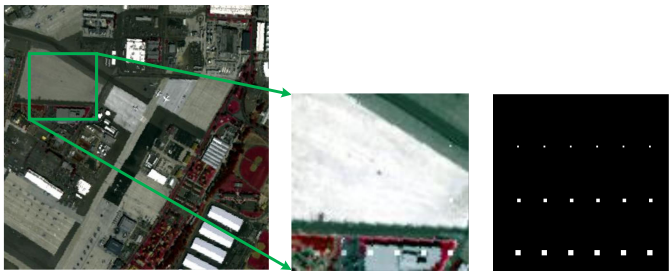


Fig. 7. Synthetic data set. (a) False-color image. (b) Ground-truth map.



Fig. 8. San Diego data set. (a) False-color image. (b) Ground-truth map.

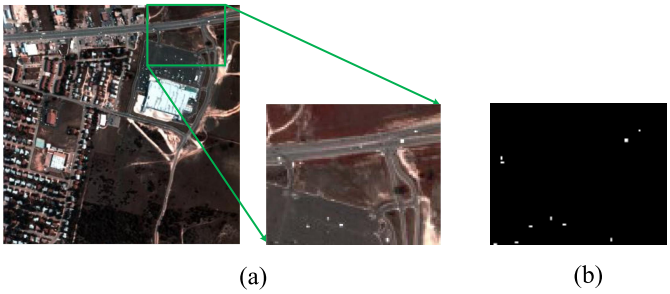


Fig. 9. Urban data set. (a) False-color image. (b) Ground-truth map.

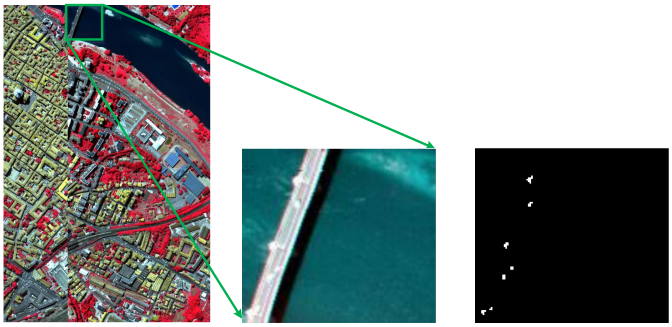


Fig. 10. Pavia data set. (a) False-color image. (b) Ground-truth map.

water vapor absorption bands. A subimage with 80 × 100 pixels in the top-right corner is intercepted for experiments. Some man-made cars and roofs containing 21 pixels totally are defined as anomaly objects. The false-color image (bands 43, 22, and 12) and the ground-truth map are illustrated in Fig. 9(a) and (b), respectively.

The third real data set is collected by the reflective optics system imaging spectrometer (ROSIS) sensor from the Pavia Center in Northern Italy. It has a spatial resolution of 1.3 m and 115 bands spanning the wavelengths from 430 to 860nm [46]; 102 bands are available after removing some useless bands. The scene in which we perform experiments has an area of 108 × 120 pixels, which contains 43 anomaly pixels representing the vehicles on the bridge and the bare soil near the bridge pier. Its false-color image (bands 102, 55, and 32) and ground-truth map are shown in Fig. 10(a) and (b), respectively.

B. Detection Performance

The detection performance of the proposed HR-JSR is compared with seven widely used anomaly detectors, including LRX, KRX, LSMAD, the low-rank and SR (LRASR) proposed in [22], the LRaSMD with cluster weighting (LSwCW) proposed in [29], the traditional neighborhood-based JSR detector (BJSRD) in [23], and SWJSR. The eight-connected region division is employed in both LSwCW and HR-JSR for segmenting the scene into different regions from the spatial domain. LSwCW directly assigns a weight value to each region by counting the number of pixels within, while, for HR-JSR, these nonoverlapping homogeneous regions are adopted as the input of JSR. Compared with BJSRD and SWJSR, HR-JSR effectively solves the fundamental problems existing in the JSR model for AD through the discarding of small regions and the homogeneous region-based JSR. In addition, the influence of noise on detection accuracy is alleviated by the incorporation of regionwise noise. All compared detectors are implemented with their optimal parameters.

The color detection maps of all compared detectors on each data set are shown in Fig. 11(a)–(h). It can be observed that KRX, LRASR, LSwCW, and SWJSR are good at highlighting anomaly targets in the scene, while LRX, LSMAD, and BJSRD can flatten the background into lower responses. JSR-based detectors, including BJSRD, SWJSR, and HR-JSR, suppress the background components on the synthetic data set almost perfectly. For all data sets, the false alarms of SWJSR are more serious than HR-JSR, especially at the edges of different materials. In general, our proposed HR-JSR achieves both satisfactory background suppression and anomaly extrusion, thus visually demonstrating its superior detection performance over others. Attributed to the successful homogeneous region-based background feature extraction and the consideration of noise, HR-JSR in particular realizes a smooth background distribution.

For quantitative comparisons, Figs. 12 and 13 provide the ROC curves and normalized background-anomaly separation maps of these compared detectors, respectively. From Fig. 12, we can see that, for all four data sets, the ROC curve of HR-JSR is almost always above that of other detectors, except for a few threshold points. Compared with other detectors, HR-JSR reaches the DR of 1 with a relatively lower FAR. The conclusion drawn from the ROC curves is consistent with the observation from the detection maps. For the background-anomaly separation maps shown in Fig. 13, it can be found that HR-JSR divides the background and anomaly boxes in a large gap with almost no overlap for all

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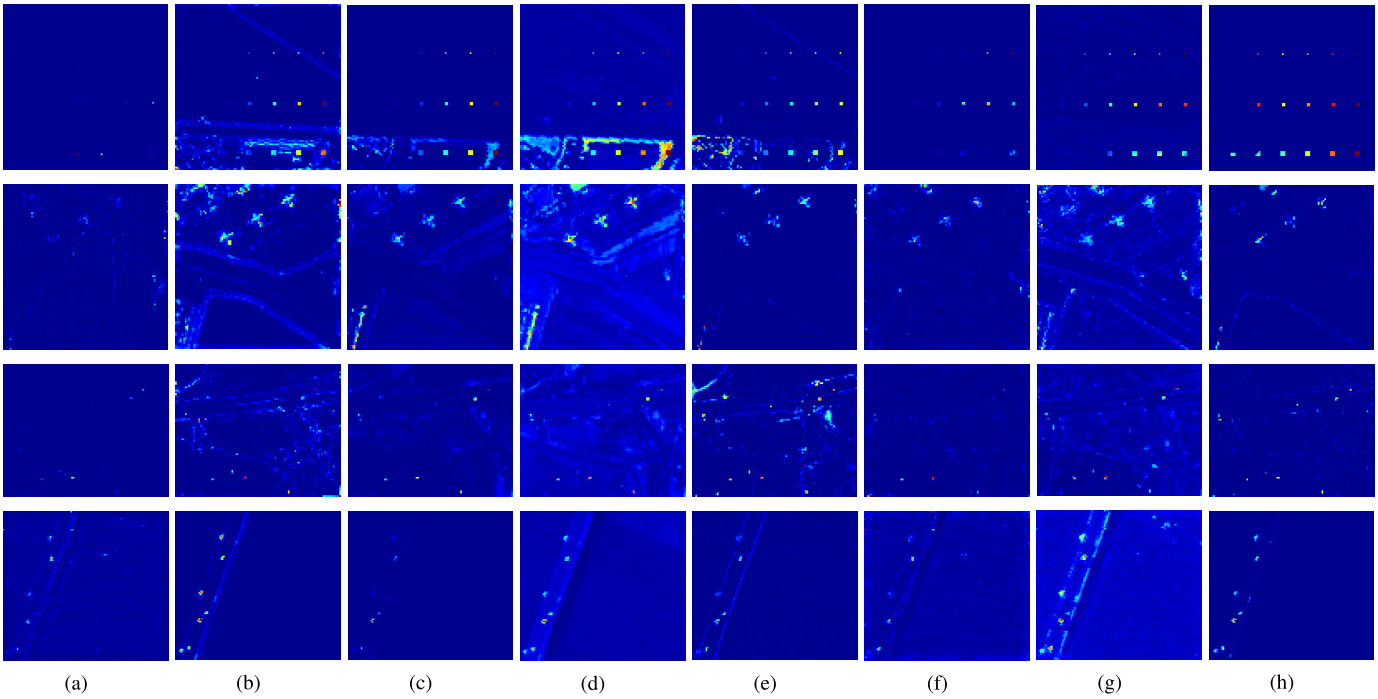


Fig. 11. Color detection maps of all compared anomaly detectors. Rows from top to bottom correspond to the four data sets, respectively. (a) LRX. (b) KRX. (c) LSMAD. (d) LRASR. (e) LSwCW. (f) BJSRD. (g) SWJSR. (h) HR-JSR.

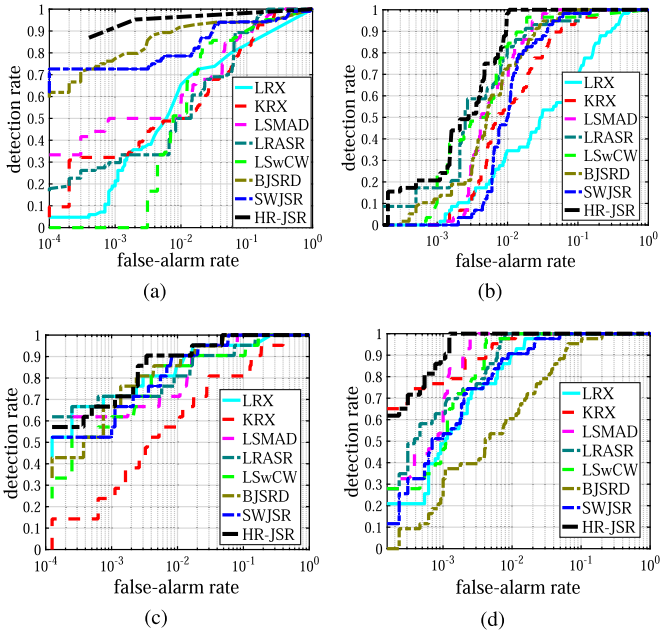


Fig. 12. ROC curves. (a) Synthetic data set. (b) San Diego data set. (c) Urban data set. (d) Pavia data set.

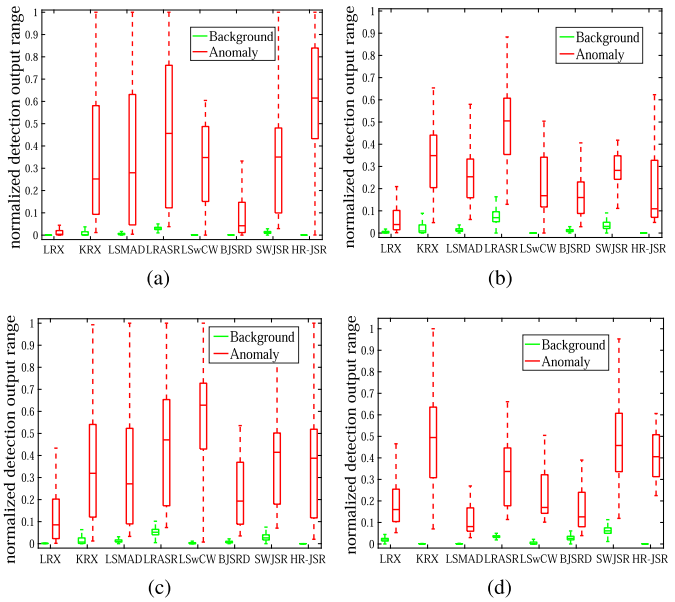


Fig. 13. Background-anomaly separation map. (a) Synthetic data set. (b) San Diego data set. (c) Urban data set. (d) Pavia data set.

four data sets, making the anomalies easy to be identified from the background.

The AUC values of all compared detectors for each data set are listed in Table I. For the Pavia data set, most of the detectors achieve satisfactory anomaly recognition, with an AUC greater than 0.99. Table I reveals that the proposed HR-JSR obtains the largest AUC value for all data sets, again quantitatively confirming its most outstanding detection ability.

Furthermore, the calculation times (in seconds) of all these detectors are given in Table II. It is obvious that LRX, LRASR, and SWJSR are not suitable for fast detection requirements. The computational burden of SWJSR is really heavy because it involves multiple complex iterations of OMP and KSVD for each pixel in the preconstruction of the background dictionary. The reason for the higher computational efficiency of HR-JSR than BJSRD is that it does not need to traverse the local spatial neighbors of each pixel in the scene. In addition, HR-JSR is more suitable for practical applications since there is no

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TABLE I
AUC VALUES OF ALL COMPARED DETECTORS FOR EACH DATA SET

AUC value	LRX	KRX	LSMAD	LRASR	LSwCW	BJSRD	SWJSR	HR-JSR
Synthetic dataset	0.8879	0.9552	0.9666	0.9671	0.9513	0.9601	0.9659	0.9762
San Diego dataset	0.9028	0.9768	0.9915	0.9893	0.9763	0.9895	0.9843	0.9927
Urban dataset	0.9882	0.9541	0.9922	0.9870	0.9852	0.9965	0.9948	0.9970
Pavia dataset	0.9967	0.9990	0.9992	0.9983	0.9984	0.9804	0.9959	0.9996

TABLE II
CALCULATION TIMES OF ALL COMPARED DETECTORS FOR EACH DATA SET

Time (s)	LRX	KRX	LSMAD	LRASR	LSwCW	BJSRD	SWJSR	HR-JSR
Synthetic dataset	39.069	4.031	14.393	44.675	17.229	27.247	779.520	21.014
San Diego dataset	43.562	6.960	15.613	47.736	16.512	51.721	893.429	19.782
Urban dataset	28.189	4.058	13.962	39.068	10.973	29.524	695.246	20.121
Pavia dataset	18.199	4.951	13.450	61.847	12.491	24.389	521.471	22.855

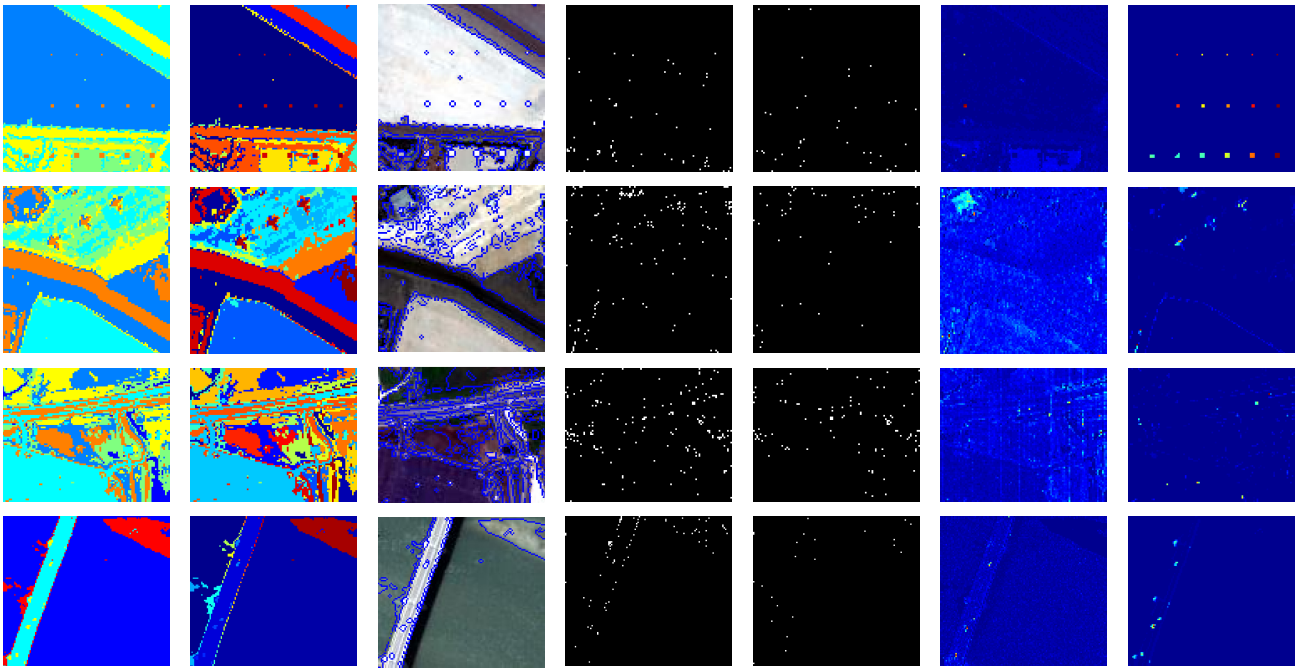


Fig. 14. Results of some staged procedures. Rows from top to bottom correspond to the four data sets, respectively. (a) Clustering map. (b) Nonoverlapping homogeneous regions. (c) Regional edges. (d) Global background bases obtained without discarding small regions. (e) Global background bases obtained after discarding small regions. (f) Detection map corresponding to (d). (g) Detection map corresponding to (e).

need to manually specify the sizes of the spatial dual windows in BJSRD.

C. Staged Procedures' Illustration

In order to provide a comprehensive understanding of our HR-JSR method, we illustrate the results of some staged procedures for all data sets. Based on the clustering maps in Fig. 14(a), the eight-connected region division operation further segments the scene from the spatial domain according to the spatial aggregation property of homogeneous materials. The obtained nonoverlapping regions of the whole scene and the corresponding regional edges are exhibited in Fig. 14(b) and (c), respectively. Due to display limitations, some regions in Fig. 14(b) appear to have similar colors visually, but, in fact, each region is painted by a unique color. Fig. 14(d) and (e), respectively, illustrates the global background bases extracted by the homogeneous region-based JSR without and with discarding small regions. Finally, the detection maps obtained by background orthogonal subspace projection with these two kinds of global background bases are exhibited in Fig. 14(f) and (g), respectively. Fig. 14(f) achieves an AUC of 0.2647, 0.6850, 0.5113, and 0.7636 for the four data sets, respectively. In contrast, the AUC of Fig. 14(g) for these data sets is 0.9762, 0.9927, 0.9970, and 0.9996, respectively, which quantitatively confirms the necessity of discarding the small regions that are most likely to correspond to the spatially aggregated anomalies in the scene.

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TABLE III
AUC AND ADDITIONAL CALCULATION TIME OF HR-JSR UNDER GLOBALWISE AND REGIONWISE NOISE SUPPRESSIONS

Datasets	Synthetic		San Diego		Urban		Pavia	
	Global	Region	Global	Region	Global	Region	Global	Region
AUC value	0.9586	0.9762	0.9916	0.9927	0.9378	0.9970	0.9993	0.9996
Additional time (s)	15.77	-	16.80	-	9.95	-	10.81	-

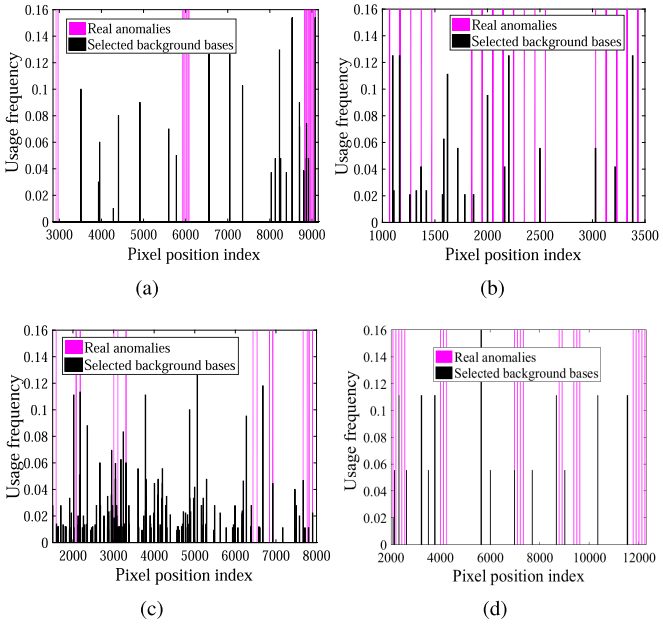


Fig. 15. Coincidence test. (a) Synthetic data set. (b) San Diego data set. (c) Urban data set. (d) Pavia data set.

To quantitatively reflect the purity of the selected background bases in Fig. 14(e), a coincidence test is performed to count the number of real anomalies mixed. As shown in Fig. 15, the spatial coordinate positions of the scene are sequentially arranged on the horizontal axis. The black bars are used to indicate the spatial positions of the selected background bases and their probabilities of being selected, that is, the normalized usage frequency. The red bars reflect the spatial positions of real anomalies in the scene. Due to display limitation, some adjacent bars seem to overlap in our view. For each subfigure, we only intercept a part of it for clear display. It can be counted that only one background base in the Urban data set coincides with the real anomaly. However, when retaining small regions, there are 10, 9, 11, and 10 background bases that coincide with real anomalies for these four data sets, respectively. Such a coincidence test reflects the purity of the selected background bases, which is actually proof of effective background feature extraction. After discarding small regions, although there may still be few anomalies in some reserved regions, they are filtered into the residual matrix so that the selected background bases will not be contaminated.

D. Effectiveness of Noise Consideration

In this section, the optimization of detection performance with the consideration of globalwise and regionwise noise suppression is compared [49]. For all data sets, the detection maps obtained from background orthogonal subspace projection without noise suppression are shown in the first row of Fig. 16. The globalwise and regionwise noise energy distribution maps are shown in the second and third rows, respectively. Then, the final detection maps of HR-JSR under globalwise and regionwise noise suppression are shown in the last two rows, respectively. The global noise is measured through the LRaSMD model, which is regarded as the most typical algorithm to separate the global noise component from an HSI. A quantitative comparison is listed in Table III, including the AUC value and the additional calculation time (in seconds) caused by noise measurement to HR-JSR. It can be concluded that compared with globalwise noise suppression, the regionwise obviously promotes a more excellent performance optimization for HR-JSR and does not introduce any additional computational burden since the homogeneous region-based JSR in Section III-B directly provides regionwise noise measurement. Therefore, regionwise noise suppression is more suitable for HR-JSR.

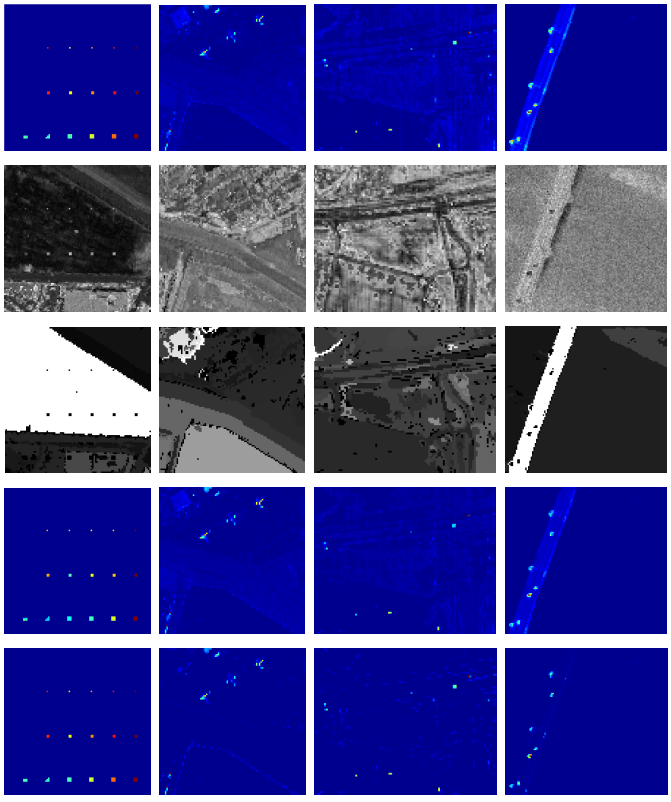


Fig. 16. Visual performance comparison of HR-JSR under globalwise and regionwise noise suppression. (a) Synthetic data set. (b) San Diego data set. (c) Urban data set. (d) Pavia data set.

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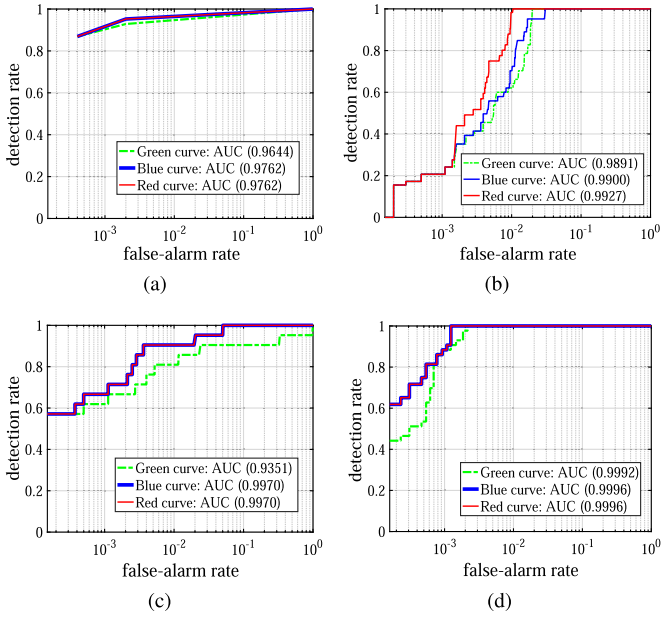


Fig. 17. ROC curves for performance comparison. The green and red curves correspond to the detection results without and with regionwise noise measurement, respectively, and the blue curve corresponds to the detection result that considers regionwise noise but does not eliminate possible anomalies. (a) Synthetic data set. (b) San Diego data set. (c) Urban data set. (d) Pavia data set.

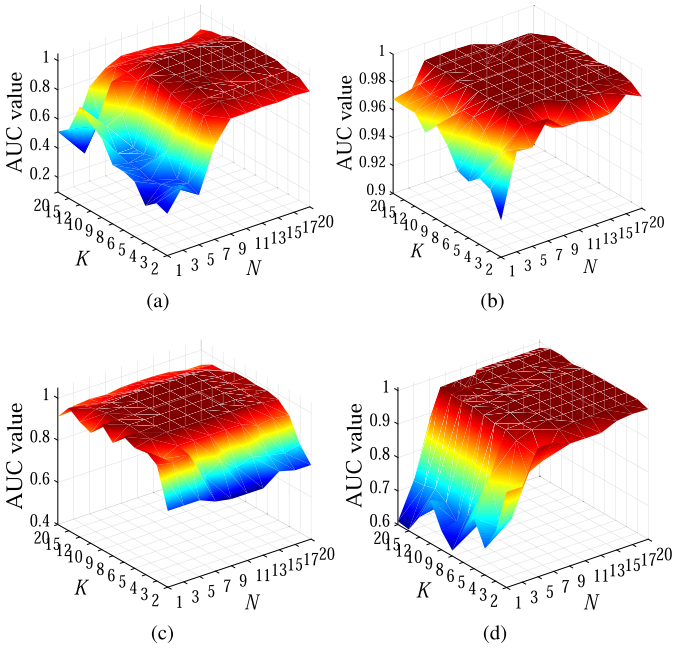


Fig. 18. AUC illustration with different combinations of K and N. (a) Synthetic data set. (b) San Diego data set. (c) Urban data set. (d) Pavia data set.

To objectively demonstrate the optimization of HR-JSR with regionwise noise suppression, Fig. 17 shows the plots of the ROC curves corresponding to the detection maps without and with regionwise noise suppression. In addition, the ROC of the detection result that considers regionwise noise but does not eliminate possible anomalies is also given in the blue curve. The AUC values are listed in brackets. It can be observed that the consideration of regionwise noise does provide quantitative optimization on the detection result, which is consistent with the visual observations from Fig. 16. For most of these data sets, eliminating possible anomalies in the residual matrix does not further improve the detection performance because the discarding of small regions has achieved the exclusion of almost all anomaly pixels in the scene.

E. Parameter Analysis

There are some parameters in HR-JSR that may influence the detection performance, including: 1) the number of clusters: K; 2) the threshold of pixel number used to define small regions: N; 3) the upper bound of the joint sparsity level: K_0; and 4) the percentage of the selected background bases for each cluster: L. Here, the robustness of HR-JSR to these parameters is proven. When we analyze a specified parameter, the other parameters remain optimal.

First, the sensitivity of the detection performance of HR-JSR to K and N is jointly analyzed since they can both affect the distribution and number of the regions for JSR. Without loss of generality, K is selected from {2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 20}, and N is set from 1 to 20 with an interval of 2. For all data sets, Fig. 18 illustrates the AUC of HR-JSR when jointly taking these two parameters into consideration.

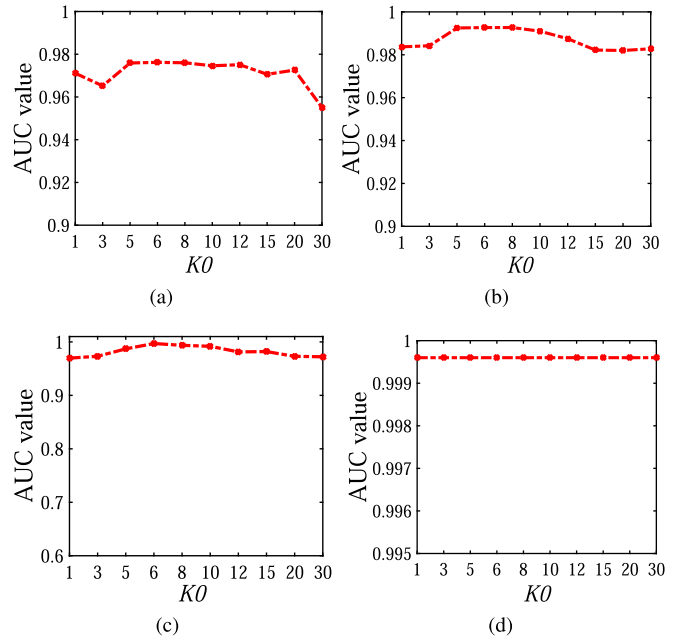


Fig. 19. AUC of HR-JSR with respect to K_0. (a) Synthetic data set. (b) San Diego data set. (c) Urban data set. (d) Pavia data set.

It can be found that most of the combinations of K and N achieve stable and satisfactory detection. When N is less than the anomaly size, the detection performance is poor. In this case, most anomalies in the scene cannot be excluded from participating in JSR, which may cause the selected background bases to be contaminated by anomalies and, eventually, lead to seriously missed detection. This phenomenon does not occur in the Urban data set, possibly due to its small anomaly size. When N is close to 20, although a few small

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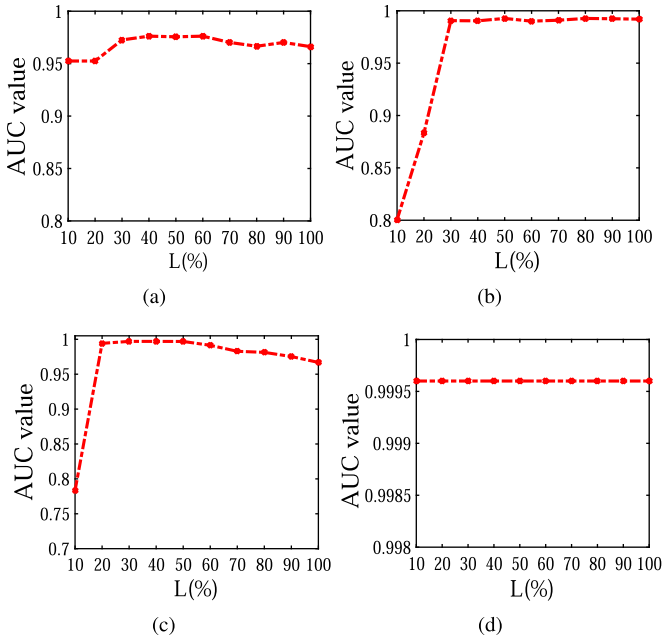


Fig. 20. AUC of HR-JSR with respect to L. (a) Synthetic data set. (b) San Diego data set. (c) Urban data set. (d) Pavia data set.

background regions may be discarded along with anomaly regions, the deterioration of detection performance is very weak since the main background information is still retained in the remaining large regions. However, through experiments, we find that the further increase in N (larger than 20) will lead to a significant decrease in AUC due to the loss of critical background features. For the Urban data set, a too small K may cause some anomaly materials to be integrated into the background regions and, eventually, lead to poor detection performance. In our experiments, K and N are set as $\{5, 9\}$, $\{5, 15\}$, $\{6, 4\}$, and $\{4, 11\}$ for these four data sets, respectively.

Then, the performance of HR-JSR with respect to the joint sparsity level K_0 is observed. K_0 is chosen from $\{1, 3, 5, 6, 8, 10, 12, 15, 20, 30\}$, and then, the variation of AUC with K_0 is shown in Fig. 19. With the increase in K_0, the detection performance of HR-JSR is quite stable, with only slight fluctuations. $K_0 = 6$ is a good choice for all the four data sets.

Finally, the influence of L on the detection performance of HR-JSR is investigated. L varies from 10% to 100% with an interval of 10%, and then, the corresponding AUC is calculated. As shown in Fig. 20, for all data sets, a small amount of background bases is already sufficient to describe the background features in each cluster. As the number of selected background bases further increases, the AUC remains unchanged or even decreases due to the introduction of some irrelevant bases. For the sake of simplicity, L is chosen as 30% in our experiments, achieving a satisfactory result for all the four data sets.

For the above four parameters, although their optimal values on various HSI data sets are not the same, they all produce satisfactory detection performance with a wide range of values for these four experimental data sets, that is, HR-JSR is quite robust to related parameters. For a new unknown data set, we can perform detection under any parameter value within this reasonable range, which can ensure an excellent detection performance to a great extent.

V. Conclusion

At present, most background feature extraction methods for hyperspectral images are limited by anomaly contamination, background incompleteness, complex spectral mixture, or loop iterations. This article provides a robust background feature extraction method through homogeneous region-based JSR for hyperspectral AD. The eight-connected region division operation based on the clustering map segments the scene from both the spectral and spatial domains. After discarding small regions, it can be largely guaranteed that the spectral items in any remaining region have very similar background spectral characteristics, thereby achieving a common sparsity pattern in each region. Through homogeneous region-based JSR, the most representative background bases for each cluster are adaptively selected and then combined into the global background bases. As a form of background feature extraction, the background base selection is superior due to the inclusion of all background categories and resistance to anomalies. It overcomes the two essential shortcomings of traditional neighborhood-based JSR and fully explores the spatial characteristics of HSIs. Furthermore, the energy deviation-based noise estimation strategy effectively alleviates the influence of noise on detection accuracy.

The experimental results on both synthetic and real-world HSI data sets fully demonstrated the superiority and effectiveness of the proposed method. However, we only considered the additive noise in the scene while ignored the multiplicative noise. In addition, the performance improvement of spatial segmentation technology and the further optimization of computational complexity are also necessary, which will be the focus of our future work.

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